



A Separation Minima Assessment for UAVs Integrating into Terminal Airspace by Monte Carlo Simulations

Linghang Meng* and Hongyang Zhang

College of Air Traffic Management

Civil Aviation University of China, Tianjin, 300300, China

Kin Huat Low*

Research Institute of Airspace Integration and Safe Operations

Civil Aviation University of China, Tianjin, 300300, China

Yifei Zhao

College of Air Traffic Management

Civil Aviation University of China, Tianjin, 300300, China

*lhmeng@cauc.edu.cn; khlow@cauc.edu.cn

Abstract

Integrating uncrewed aerial vehicles (UAVs) into national airspace systems introduces human-machine interaction paradigms, amplifies systemic safety risks, and challenges the adaptability of existing civil aviation regulatory frameworks. To address the safety risks, determining the appropriate separation minima is a key measure. This study focuses on the integration of UAVs within terminal airspace, examining three mitigation phases: strategic, tactical, and collision avoidance. It proposes a trajectory conformity-based collision risk assessment model that quantifies failure probabilities across interconnected subsystems during potential UAV/MAV encounters. From the perspective of regulatory authorities, a quick-estimated equivalent-target-safety-level (ELOS) metric has been developed to reasonably assess the separation requirements for UAV operations, balancing the kinetic energy characteristics of UAVs with the socially acceptable level of safety. Finally, the Monte Carlo simulation methodology was systematically implemented to conduct a comprehensive safety assessment of multiple traffic separations under varying traffic density, UAV cruising speeds, and UAV-to-MAV ratios, to determine appropriate horizontal and vertical safety separation minima for different operational scenarios in terminal airspace.

I. Introduction

The integration of unmanned aerial vehicles (UAVs) into terminal control areas (TCAs) introduces distinct operational risks arising from heterogeneity in avionics systems, flight dynamics, and human-machine interfaces [1]. From the perspective of the aviation administration and Air Traffic Management (ATM) agents, the paramount rule for integrating UAVs into the TMA shall not compromise the current system's safety level while complying with the existing flight rules to the extent possible [2]. Furthermore, the safety level of the current MAV's separation minima, adapted to the paradigm of UAV-MAV integrative operation, shall be rigorously assessed. However, conventional MAV-MAV collision models do not demonstrate adequate representational fidelity for the integrative UAV-MAV operational environment. Firstly, air traffic control (ATC) instructions are transferred to UAVs via combined VHF/SATCOM and Command-and-Control (C2) links, thereby increasing transaction time and the likelihood of

communication failure [3]. Secondly, a single remote pilot must simultaneously monitor and operate multiple UAVs, which in turn introduces a higher risk of human error [4]. Thirdly, while UAVs possess greater maneuverability potential than MAVs, their reliance on C2 links introduces critical latency [5]. Consequently, UAVs should routinely yield right-of-way to MAVs, while MAVs should accommodate emergency avoidance maneuvers initiated by UAVs. These three operational distinctions necessitate revisions to conventional aircraft collision models and separation minima assessment frameworks to appropriately address the unique integration challenges presented by UAVs in TMA. Moreover, the fatality risk tolerances following UAV-MAV collisions exceed those for MAV collisions alone because there is no passenger onboard, and the UAV has less kinetic energy. ICAO recommends that the integration of UAVs into civil airspace shall maintain the established Target Level of Safety (TLOS) for MAV operations without degradation [6]. This safety equivalence, commonly referred to as the Equivalent Level of Safety (ELOS) principle among national aviation authorities, currently lacks a universally accepted quantitative framework for precise operationalization.

This paper concerns the assessment of UAV operational separation minima in terminal airspace and is organized as follows: firstly, we provide a review of the literature in Section II; in Section III, we propose the methodology to determine ELOS and establish the collision risk model for the entire conflict process. Moreover, we evaluate the separation minima using a Monte Carlo simulation in Section IV. Then, Section V validates the method in determining separation minima. Finally, we summarize the findings in Section VI.

II. Literature Review

The formulation of separation minima for UAVs faces three main challenges: determining the ELOS, establishing collision risk models, and developing effective evaluation methods for separation minima.

A. ELOS

The primary principle established by ICAO for UAV integration mandates that the current TLOS standards for piloted aircraft shall not undergo degradation. The TLOS defined by ICAO represents the minimum safety objective that must be ensured during operations, and it is expressed in terms of the socially acceptable frequency of fatal events [6], [7]. TLOS can be based on the number of accidents per flight hour or operation, often using historical fatal accident data for derivation [8]. The initial TLOS in civil aviation was set at one landing accident in 10 million landings, which was used to approve an automatic "all-weather" landing system in the 1950s. The TLOS target of 1×10^{-7} is widely recognized as a key overarching objective in commercial aviation by the International Civil Aviation Organization (ICAO) and must be decomposed into each critical subsystem [9]. Specifically, the North Atlantic Systems Planning Group (NATSPG) recommends that collision risk should account for 10% of total operational risks, with each spatial dimension of the separation risk further allocated one-third of this subset [6]. Various TLOS have been developed in the past centuries by researchers, research panels, and aviation authorities [10].

Owing to substantial disparities between UAVs and MAVs in terms of airborne equipment, innovative human-machine interfaces, and affected stakeholders, it is critical to redefine the ELOS for UAV operations to establish appropriate safety standards [11]. The initial approach to determining the ELOS focused on the likelihood of ground fatalities resulting from failures or collisions leading to crashes [12]. The criterion established by the European Union Aviation Safety Agency (EASA) that defines the ELOS for UAV operation is that the probability of causing harm to people and property on the ground must be no greater than that of MAV [13], [14]. JARUS proposed an equation to develop the ELOS from the current general aviation safety level by considering ground population density [17]. However, the assumption that every crash event would necessarily result in a casualty is inaccurate, as the analysis oversimplifies by neglecting the energy-related impacts of UAVs.

B. Collision Risk Model

The collision risk models for traditional MAVs have been rigorously developed and established by leading research institutions such as MIT in [15][16]. The most distinguished model is the Reich model (1966), which incorporates aircraft encounter frequency and three-dimensional positional uncertainty as key parameters to evaluate the operational separation minimum for parallel air routes in the North Atlantic airspace [18]-[20]. However, the Reich model does not account for key collision prevention mitigations, including air traffic control procedures, Traffic Collision Avoidance System (TCAS), and pilot interventions. Booker analyzed the root causes of collision risk and introduced the concepts of Position Integrity and Reasonable Intent [21]. Building on this insight, he proposed the Event model [22], an enhancement of the Reich model, which integrates ATC failure probabilities across operational stages and analyzes dynamic collision risk throughout conflict processes. Endoh et al. developed the gas model to calculate the collision rate by analogizing aircraft motion to particle dynamics, assuming that aircraft are

uniformly distributed and move straight through the designated airspace with random velocities [23]. However, its assumptions of uniformly distributed aircraft and neglect of collision avoidance measures and ATC mitigation limit its applicability to unstructured airspace. Building upon these foundational models, contemporary researchers have integrated methodologies from systems engineering, stochastic processes, and graph networks to systematically analyze risk factors across the human-machine-environment-management framework during conflict scenarios [24]-[26]. These approaches enable more rigorous and holistic collision risk assessments across diverse operational modes.

Most current collision risk models for UAVs are improvements upon traditional models, primarily by refining the collision box to more representative geometries such as frustums, cylinders, or composite structures, to more accurately capture the physical contours of UAVs [27], [28]. However, these models primarily focus on non-integrative UAV operations and do not adequately capture human failure modes resulting from emerging human-machine interfaces in MAV-UAV integration. In controlled airspace where UAVs operate alongside MAVs, UAVs are generally categorized as either cooperative or non-cooperative [29]-[31]. Non-cooperative UAVs pose significant risks to the safe operation of MAVs due to the uncertainty of their flight status. To address this problem, many researchers have employed stochastic processes [32], [33], neural networks [34], and statistical methods [35] to estimate the states of non-cooperative UAVs and subsequently assess their collision risk with MAVs during intrusions. For a single air traffic controller managing integrative operations involving cooperative UAVs and MAVs, several factors introduce operational safety risks. These factors encompass controller response and communication delays, the accuracy of ATC instructions, and the appropriateness of pilot actions. Shively et al. [37] from NASA defined the time required for each stage of the UAV coordination process as Measure Response (MR), and provided the mean and variance of MR times for different types of instructions. Carbó et al. [38] conducted a quantitative assessment of UAV collision risk by analyzing the failure probability of each mitigation measure designed to prevent collisions, based on the conflict mitigation process proposed by the BUBBLES.

C. Separation Assessment

The fundamental requirement for UAVs integrating into manned airspace is that they must not compromise the TLOS of existing air operations. There is also an expectation that current flight rules and separation minima for MAVs remain unchanged to prevent the introduction of new human-error risks. Therefore, safety validation of existing separation minima is required to verify the preservation of equivalent TLOS during MAV-UAV integration. There are currently three main types of separation minima for UAVs: distance-based, time-based, and combined time-distance.

The distance-based method defines a safety buffer zone around the UAV. If another UAV intrudes into this zone, the separation is considered lost [15], [39]. Johnson et al. [40] cited UAV autonomy spacing models in RTCA SC-228 to derive the UAV autonomy separation in Class E airspace, providing technical support for developing the minimum operational performance standards (MOPS) for UAV DAA systems. However, this approach does not account for the relative velocity between UAVs. The time-based method addresses this limitation by accounting for relative speed and using the time to the closest point of approach (CPA) as the criterion for maintaining separation [31]. Mullins et al. [41] investigated the dynamic spacing for UAV-based DAA systems using a time-based Well-Clear threshold, accounting for UAV dynamics. Nevertheless, this method lacks an intuitive visual representation. To overcome these limitations, the combined time-distance method integrates the advantages of both approaches. It first computes the time it takes for the intruding aircraft to reach the CPA and then converts this time into a separation distance based on the Closing rate, thereby providing a more comprehensive and dynamic evaluation of UAV separation safety. Munoz [42] et al., based on the resolution advisory logic of TCAS II (Traffic Alert and Collision Avoidance System), calculated the "Well Clear" threshold by considering the relative motion between the intruder and the unmanned aircraft system, quantitatively defining the minimum horizontal and vertical separation distances in terms of both time and distance, and subsequently determining the temporal separation according to the closest point of approach. Johnson et al. [43] analyzed the differences between unmanned and manned aircraft in existing Class E airspace, proposed three separation definition schemes, and established a simulation environment incorporating unmanned aircraft system models, flight profiles, and historical radar data to validate the proposed schemes. Meng et al. [44] developed a collision risk model based on the random intrusion positions and headings of unmanned aircraft, accounting for the delay time of collision-avoidance maneuvers, and calculated three-dimensional "Well Clear" distances in free airspace.

In summary, current research on collision risk assessment and calibration of separation minima in TMA for UAV-MAV integrative operations remains limited. The world's first regular operation of UAV-MAV integration began at the Zigong UAV Flight Demonstration Base in Sichuan, China, on April 1, 2024. The UAV-MAV integrative operation separation minima incorporated in Zigong TMA are 10km/600m, which are significantly larger than those for conventional MAV operation. Based on the operational framework for Zigong's UAV-MAV integrated operations,

this study develops a comprehensive separation assessment methodology that quantifies UAV positional integrity, integrates intent-based ATC adjustments, and establishes safety-critical separation standards for mixed-airspace operations.

III. Methodology

A. Quick Determination of ELOS

Determining the TLOS for the UAV-MAV integrative operation in terminal areas is a rigorous systems engineering problem. Currently, there are no unified standards for integration rules or equipment requirements for UAV operations in the TMA. Therefore, it is difficult to analyze in detail the casualties caused by failures of individual UAV subsystems. This research follows the guiding principle that the integration of UAVs into TMA must not compromise the TLOS established for conventional MAV operations [6], [7]. To this end, the TLOS for UAV operations is quantitatively assessed by examining operational differences between UAVs and MAVs, with a particular focus on kinetic energy in Mid-Air collisions and societal acceptance of the fatality risks. Under the premise that UAVs and MAVs exhibit comparable kinetic energy levels in TMA, we define a proportionality factor η relating TLOS to ELOS, representing the ratio of fatality probabilities from UAV-MAV Mid-Air collisions to those from conventional MAV operations. It should be emphasized that both ELOS and TLOS in this study represent thresholds for acceptable Mid-Air collision risk within TMAs, not overarching system safety objectives. Therefore,

$$ELOS = \eta \times TLOS \quad (1)$$

and

$$\eta = \eta_k \times \eta_s \quad (2)$$

where η_k denotes the kinetic ratio between UAV-MAV and MAV-MAV, η_s denotes the public acceptance level ratio to casualties. Moreover,

$$\eta_k = \frac{M(2V)^2}{m(V+v)^2} \quad (3)$$

where M and m are the expectation of mass of MAV and UAV, V and v are the expectation of speed of MAV and UAV.

Due to the number of casualties caused by an accident being closely related to the public's level of acceptability, and according to China's acceptable societal risk standard, there is an exponential relationship between the number of casualties and the socially acceptable frequency of accidents, i.e.,

$$\eta_s = 10^{\frac{P}{p}} \quad (4)$$

where P and p denote the number of passengers affected in MAV-MAV collisions and MAV-UAV collisions, respectively.

Thus, a quick determination of the ELOS is given by,

$$ELOS = \frac{M(2V)^2}{m(V+v)^2} \times 10^{\frac{P}{p}} \times TLOS \quad (5)$$

It should be highlighted that Eq. (5) is served as a model for quick determination. The model requires further examination and a careful validation as a future work.

B. Collision Risk Modeling

Within the current air traffic management (ATM) paradigm in TMA, aircraft collision prevention is ensured through a four-layer risk mitigation framework: (i) decoupling the air traffic flow through Airspace Management (ASM) on strategic level, (ii)balancing the traffic demand and airspace capacity via Air Traffic Flow Management (ATFM) on pre-tactical level, (iii) detecting and resolving conflicts via Air Traffic Control (ATC) on tactical level, (iv) avoiding collisions via airborne collision avoidance systems on real time level.

The baseline encounter rate $N_{\text{encounter}}$ is primarily governed by Pre-Tactical ATFM (PTCA) measures. However, the inherent stochasticity of aircraft arrivals introduces probability $P_{\text{TCA|PTCA_fail}}$ that converts a portion of encounters into actual conflicts. ATC controllers identify conflicts by assessing whether two aircraft encountering each other infringe both vertical and horizontal separation minima simultaneously at the Closest Point of Approach (CPA), based on their spatial and temporal convergence. For any encounter declared a conflict, the controller issues Tactical Collision Avoidance (TCA) instructions, typically adjusting speed, altering heading, or changing altitude, to the

aircraft lacking right-of-way to ensure the Well-Clear (WC) threshold is not infringed. Under current flight rules in Zigong TMA, MAVs generally retain priority on regular conditions, while UAVs assume the right-of-way during emergency collision avoidance maneuvers. ATC failures primarily stem from excessive workload, compounded by the need to resolve concurrent conflicts and the inherent complexity of dynamic traffic environments. Currently, no rigorously validated model exists to quantify the relationship between the intrinsic complexity of air traffic and ATC failure rates. As a pragmatic alternative, we simplify the ATC failure process by modeling it as a time-critical function: comparing the total delay in the controller-remote pilot Conflict Detection-and-Resolution (CD&R) cycle against the remaining time for a conflicting aircraft to reach CPA. As traffic complexity and conflict density increase, the handling period lengthens, which delays lower-priority aircraft to the point of violating the WC threshold, forcing Emergent Collision Avoidance (ECA) to resolve the situation.

Finally, the failure of Detect-and-Avoid (DAA), whether due to insufficient TCA maneuvering time, DAA malfunction, or unresolved NMAC, will escalate the encounter from the ECA past the Near Mid-Air Collision (NMAC) boundary with the probability $P_{\text{NMAC|ECA_fail}}$. At this stage, the Mid-Air Collision (MAC) probability continuously increases without further mitigation measures till to the update CPA with the probability $P_{\text{overlap_CPA}}$ at the CPA.

The sequential failures of the four-layer safety barriers can be characterized by the Swiss Cheese Model [16], where a collision occurs only if all defense-in-depth protective layers are sequentially breached (as illustrated in Fig. 1). Therefore, the collision risk during a given period can be formulated as

$$N_{\text{collision}} = 2N_{\text{encounter}}P_{\text{TCA|PTCA_fail}}P_{\text{ECA|TCA_fail}}P_{\text{NMAC|ECA_fail}}P_{\text{overlap_CPA}} \quad (6)$$

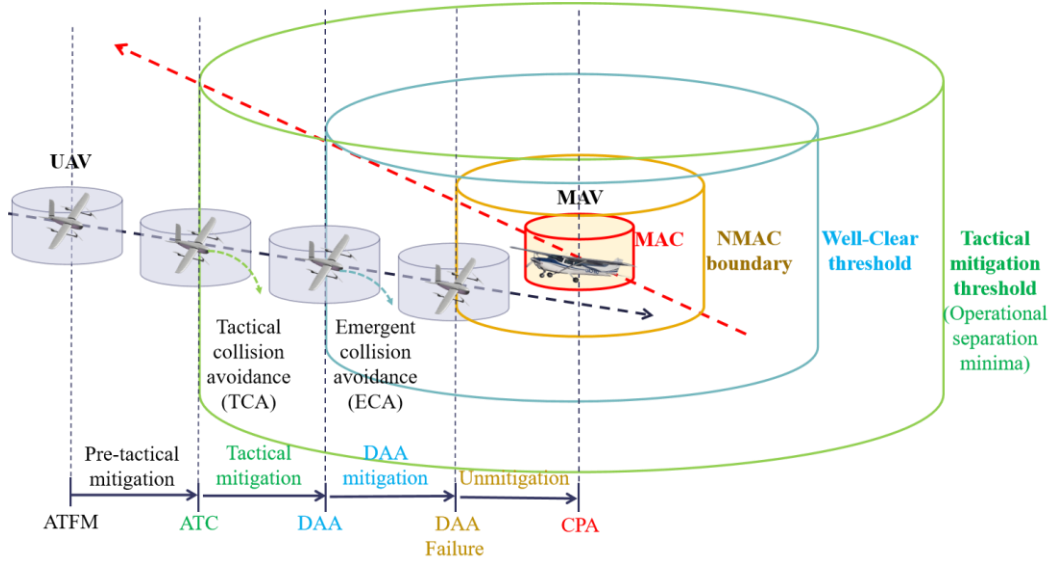


Fig. 1 Illustrated aircraft encountering process

The following process determines each element in Eqs. (6) and (7).

i. Encounter and conflict rates

Although UAVs and MAVs within terminal airspace operate under pre-filed flight plans, real-world disruptions, such as adverse weather, distributed ATC interventions, and operational contingencies, inevitably introduce stochastic deviations in their actual arrival times. Empirical evidence suggests that interarrival times between aircraft at a single fix typically follow the Poisson distributions [48]. Assume the arrival rate is λ , and it follows a Poisson distribution.

$$f = \lambda e^{-\lambda t} \quad (7)$$

Let τ denote the threshold between aircraft considered to be in conflict; therefore, for an encounter event, the conflict probability is

$$p(t < \tau) = 1 - e^{-\lambda \tau} \quad (8)$$

Therefore, the encounter counts during $[0, T]$, as a Poisson process with the arrival rate λ , can be decomposed into two independent Poisson sub-processes, the conflict process and the non-conflict process, using the Poisson Splitting Theorem [48]. For the conflict process during $[0, T]$, the conflict count m follows a Poisson process with the expectation $\lambda T(1 - e^{-\lambda \tau})$, i.e.,

$$P_{\text{TC|SC}}(N_{\text{conflict}} = m | N_{\text{encounter}}) = \frac{[\lambda T(1 - e^{-\lambda \tau})]^m}{m!} \exp[-\lambda T(1 - e^{-\lambda \tau})], m=1,2,\dots,N_{\text{encounter}} \quad (9)$$

ii. Tactical mitigation failure

Tactical mitigation involves air traffic controllers issuing instructions to UAV remote pilots to perform TCA maneuvers when a conflict is detected. This intervention prevents further separation degradation from reaching the Well-Clear threshold, thus avoiding DAA system alerts. With the assumption that both controller and remote pilot are competent, i.e., their human error failures are neglected, the tactical mitigation failure can result from either the controller's decision-making delay due to the high conflict rate or complexity or the remote pilot's reaction delay in scanning display to identify the correct UAV, to aggravate the conflict into breaking the Well-Clear (WC) threshold.

Currently, no research has explicitly quantified the relationship between the ATC decision-making delay and both encounter counts and conflict complexity. We simplify the controller's decision-making process by subdividing it into a sequential series of elementary mental calculations, including conflict detection, sorting, and resolution. The ATC decision-making delay can be approximated by linearly summing the durations of the aforementioned cognitive processes. Based on our testing on the competent controllers in Zigong TMA, each elementary mental calculation adheres to a Gaussian distribution $\tau \sim N(2.5, 1.5)$ in seconds, therefore the total delay for n concurrent encounters follows $t_{ATC} \sim N(3n\mu, \sqrt{3n}\sigma)$, i.e., $t_{ATC} \sim N(7.5n, 2.6\sqrt{n})$, which is very close to the empirical results in Ref. [36]. In comparison, the remote pilot experiences response delay only when scanning the ground station screen and pinpointing the correct UAV to execute a collision avoidance (CA) maneuver, which depends on the number of UAVs a pilot simultaneously monitors and operates. The most conservative time of ATC response and reaction is $t_{pilot} \sim N(6.9, 2.45)$.

Therefore, the probability of tactical mitigation failure is simulated using the Monte Carlo approach, which generates both ATC decision-making delays and remote pilot response delays during each encounter.

iii. Well-clear threshold

The Well-Clear (WC) threshold represents the minimum separation distance at which a UAV remote pilot must initiate Emergent Collision Avoidance (ECA) in response to a DAA alert, ensuring timely maneuver execution to prevent NMAC volume penetration. This threshold includes system latency from DAA operations and UAV avoidance maneuvers after mitigation fails.

$$d_{WC} = d_{\text{maneuver}} + d_{NMAC} + v_{\text{closure_rate}} t_{UAV_delay} \quad (10)$$

In this framework, d_{maneuver} defines the minimum required separation distance for safe UAV collision avoidance maneuvers; $v_{\text{closure_rate}}$ denotes the relative closure rate between the two aircraft which defines the dynamic WC threshold; d_{NMAC} denotes the NMAC volume set based on the safety-critical requirement of the specific operational scenario; t_{UAV_delay} incorporates the total DAA system latency, including detection, processing, and response delays.

a) d_{maneuver}

In lieu of RTCA DO-365 [44] and EUROCAE ED-229 [45], there are three types of ECA maneuvers as turning, climbing/descending and decelerating.

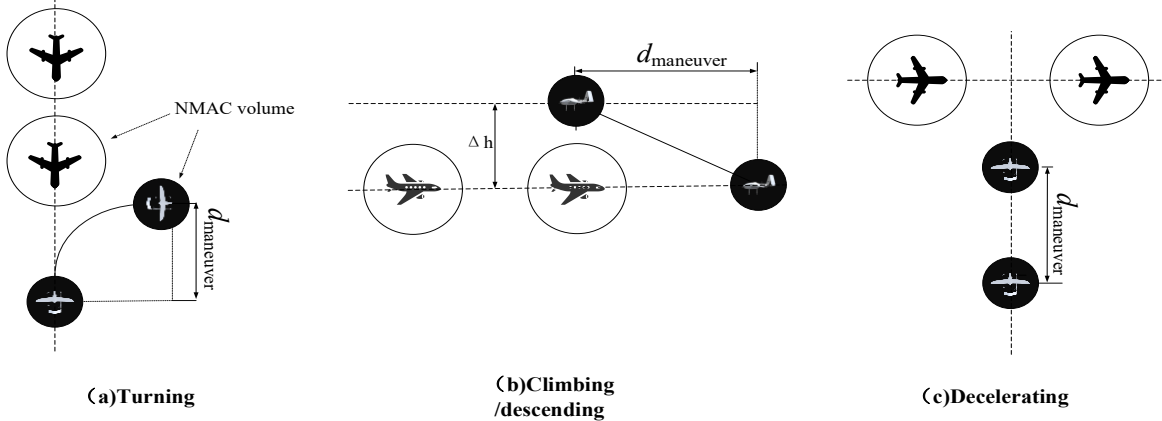


Fig. 2 Three types of ECA maneuvers triggered by DAA

For the ECA of turning, the UAV deviates from its intended trajectory by making a heading alteration with an aggressive bank angle to avoid spatial overlap of the two aircraft's NMAC volumes. The ECA distance d_{maneuver} is the turning radius.

$$d_{\text{maneuver}} = \frac{v_{uav}^2}{g\sqrt{(1/\cos\gamma)^2 - 1}} \quad (11)$$

where g is gravitational acceleration, γ is the bank angle, and v_{uav} is the turning velocity of the UAV.

For the ECA of climbing or descending, the UAV executes a vertical maneuver at its nominal climb/descent rate to establish safe separation, ensuring both aircraft's NMAC volumes remain non-overlapping. The ECA distance $d_{maneuver}$ is calculated by

$$d_{maneuver} = v_{uav} \frac{\Delta h}{R_C} \text{ or } d_{maneuver} = v_{uav} \frac{\Delta h}{R_D} \quad (12)$$

where Δh is the total vertical separation buffer calculated as the sum of both aircraft's NMAC volumes in the vertical dimension, and R_D represents the rate of descent.

For the ECA of decelerating, the UAV rapidly decelerates at maximum allowable performance to reach a safe maneuvering speed, ensuring spatial separation by preventing overlap between their NMAC volumes. The ECA distance $d_{maneuver}$ is calculated by

$$d_{maneuver} = \frac{v_{uav_final}^2 - v_{uav_initial}^2}{2a} \quad (13)$$

where $v_{uav_initial}$ is the initial speed of UAV at the ECA moment, v_{uav_final} is the final speed after finishing the deceleration process and a is the deceleration of UAV.

iv. NMAC volume

The NMAC defines the critical proximity within which the risk of an actual collision between aircraft becomes significant. The NMAC volume is a 3-dimensional protected space around an aircraft where the conditional probability of an actual MAC event exceeds the predefined Severity of Loss of Well-Clear (SLoWC) threshold. A higher SLoWC necessitates a larger NMAC volume, thereby increasing both the WC threshold and the minimum safe separation. This scaling is nonlinear and depends on the relative position and velocity, resulting in quite different shapes and sizes of NMAC volumes. For simplicity, this research conservatively assumes the miss distance at CPA to be zero when deriving the NMAC volume size, as illustrated in Figure 3.

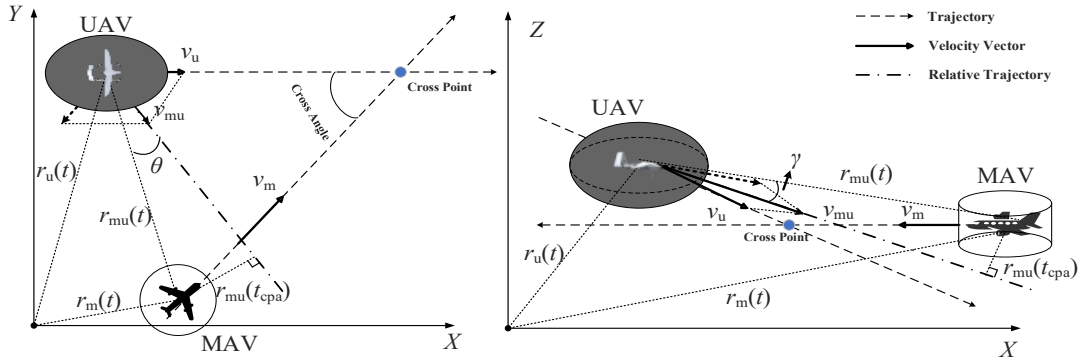


Fig. 3 The horizontal and vertical encounter process between UAVs and MAVs

Let R_{uav} and R_{mav} be the radii of the UAV and MAV's cylindrical collision boxes; H_{uav} and H_{mav} be the half-heights of the UAV and MAV's cylindrical collision boxes. The nominal position and actual position vectors of the UAV and MAV at t are represented by $\mathbf{r}_{uav}(t)$, $\mathbf{r}_{mav}(t)$, $\mathbf{r}_{d_uav}(t)$ and $\mathbf{r}_{d_mav}(t)$, respectively. And let the initial position of the UAV and the MAV at t_0 be $\mathbf{r}_{uav}(t_0)$, $\mathbf{r}_{mav}(t_0)$. Since the aircraft strictly follow predefined flight procedures in structured terminal airspace, it is assumed that the aircraft's horizontal and vertical tracking errors are statistically independent. Therefore, the nominal and actual relative position vectors are $\mathbf{r}_{mu}(t)$ and $\mathbf{r}_{\bar{mu}}(t)$ are

$$\mathbf{r}_{mu}(t) = \mathbf{r}_{uav}(t) - \mathbf{r}_{mav}(t) \quad (14)$$

$$\mathbf{r}_{\bar{mu}}(t) = \mathbf{r}_{d_uav}(t) - \mathbf{r}_{d_mav}(t) \quad (15)$$

and the relative velocity vector $\mathbf{v}_{mu}$ is

$$\mathbf{v}_{mu} = \mathbf{v}_{uav} - \mathbf{v}_{mav} \quad (16)$$

where, $\mathbf{v}_{uav}$ and $\mathbf{v}_{mav}$ denote the velocity vectors of UAV and MAV.

Let the trajectory conformities of UAV and MAV, representing the total system tracking error [44], be $\mathbf{T}_{c_uav}$ and $\mathbf{T}_{c_mav}$, which follow the three-dimensional Gaussian distributions, then the relative position vector of the UAV and MAV can then be obtained from

$$\mathbf{r}_{mu}(t) = \mathbf{r}_{d_uav}(t) - \mathbf{r}_{d_mav}(t) + \mathbf{R}_{uav}\mathbf{T}_{c_uav} - \mathbf{R}_m\mathbf{T}_{c_mav} \quad (17)$$

in which

$$\mathbf{r}_{uav}(t) = \mathbf{r}_{d_uav}(t) + \mathbf{T}_{c_uav} \quad (18)$$

$$\mathbf{r}_{mav}(t) = \mathbf{r}_{d_mav}(t) + \mathbf{T}_{c_mav} \quad (19)$$

where $\mathbf{T}_{c,uav} \sim N_3(\boldsymbol{\mu}_{uav}, \boldsymbol{\Sigma}_{uav})$ and $\mathbf{T}_{c,mav} \sim N_3(\boldsymbol{\mu}_{mav}, \boldsymbol{\Sigma}_{mav})$. Note $\mathbf{R}_{uav}$ and $\mathbf{R}_{mav}$ are the rotation matrices of UAV and MAV from the body coordinate system to the ground coordinate system, respectively. Since the horizontal and vertical velocities are independent, $\mathbf{R}_{uav}$ and $\mathbf{R}_{mav}$ can be expressed as

$$\mathbf{R}_{uav} = \begin{pmatrix} \cos\alpha_{uav} & -\sin\alpha_{uav} & 0 \\ \sin\alpha_{uav} & \cos\alpha_{uav} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (20)$$

$$\mathbf{R}_{mav} = \begin{pmatrix} \cos\alpha_{mav} & -\sin\alpha_{mav} & 0 \\ \sin\alpha_{mav} & \cos\alpha_{mav} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (21)$$

where α_{uav} and α_{mav} denote the angles between the projection of the along-cross direction on the horizontal plane and the X-axis of the body coordinate system of UAV and MAV, respectively. Therefore, the relative position vector of UAV and MAV also follows the Gaussian distribution, i.e.,

$$\mathbf{r}_{mu}(t) \sim N_3(\mathbf{r}_{\overline{mu}}(t) + \mathbf{R}_{uav}\boldsymbol{\mu}_{uav} - \mathbf{R}_{mav}\boldsymbol{\mu}_{mav}, \mathbf{R}_{uav}\boldsymbol{\Sigma}_{uav}\mathbf{R}_{uav}^T + \mathbf{R}_{mav}\boldsymbol{\Sigma}_{mav}\mathbf{R}_{mav}^T) \quad (22)$$

The collision risk probability between UAV and MAV at each time step t can then be expressed as

$$P(t) = \iint_{\substack{(\Delta x(t))^2 + (\Delta y(t))^2 \\ \leq (R_{uav} + R_{mav})^2}} \mathbf{r}_{mu}(\Delta x(t), \Delta y(t)) dx dy \int_{\substack{-(H_{uav} + H_{mav}) \leq \\ \Delta z(t) \leq (H_{uav} + H_{mav})}} \mathbf{r}_{mu}(\Delta z(t)) dz \quad (23)$$

where $\Delta x(t)$, $\Delta y(t)$ and $\Delta z(t)$ represent the three-dimensional relative distance between UAV and MAV at each time step t .

The aviation sector conventionally adopts $P(\text{MAC}|\text{NMAC})=5\%$ as the standard threshold probability for small UAV operations to maintain operational efficiency and safety [47]. However, given the heightened risk sensitivity associated with UAV integration into civil terminal airspace, this study implements a more conservative criterion of $P(\text{MAC}|\text{NMAC})=0.1\%$. This conditional collision probability $P(\text{MAC}|\text{NMAC})$ exhibits a continuous monotonic increase as aircraft proximity decreases. The NMAC distance d_{NMAC} for a given collision scenario is operationally defined as the relative separation at which $P(\text{NMAC}|\text{ECA_fail})$ reaches 0.1% at time step t_{NMAC} .

$$d_{\text{NMAC}} = \|\mathbf{r}_u(t_0) - \mathbf{r}_m(t_0)\| - \|\mathbf{v}_{mu}\| * t_{\text{NMAC}} \quad (24)$$

IV. Simulation and Results

A. Scenarios

This study employs a Monte Carlo simulation framework to model the stochastic dynamics of cascading failures in the ATC system. Key factors include encounter rates, ATC operational failures, DAA system failures, and trajectory compliance deviations. By statistically aggregating collision risk across a sufficiently large number of simulation runs, this approach ensures comprehensive coverage of the sample space, providing a robust probabilistic assessment of operational safety under uncertainty.

i. Random variables

Encounter pairs are generated according to a Poisson distribution, where the rate parameter λ (encounter pairs per hour) represents traffic density, and the mix ratio ζ defines the proportion of UAVs to MAVs. An ATC failure event occurs when the total accumulated delay from handling multiple concurrent conflicts exceeds a threshold, causing at least one aircraft to violate the WC threshold. These delays are modeled as Gaussian random variables to capture realistic variations in ATC response. The DAA failure rate in this study incorporates not only hardware malfunctions but also the residual risk of a near-midair collision (NMAC) event when the DAA is functional. According to the RTCA MOPS [49], the total failure probability of DAA must be less than 1×10^{-7} . Meanwhile, ASTM F3442/F3442M stipulates that the probability of an NMAC occurring under normal DAA operation should not exceed 0.18 [49]. Given these constraints, we assume DAA system failure is negligible, focusing instead on scenarios where the DAA is functional but still fails to prevent an NMAC event. Accordingly, whenever a UAV breaches the WC threshold, we implement a Roulette Wheel Selection approach, randomly sampling a value between 0 and 1 to determine if an NMAC occurs, with a fixed probability of 0.18 representing the DAA's residual risk.

ii. Encounter geometries

Different encounter scenarios may result in significantly varying risk levels. However, ATC operations rely only on standardized horizontal and vertical separation minima for the purpose of mitigating workload and minimizing human-factor errors. To systematically evaluate these risks, we construct a generalized encounter scenario in which two aircraft converge at angles ranging from 30° to 150° with 30° increments across different flight levels,

encompassing typical TMA conflict geometries. The simulation enables a specific TMA to derive context-dependent separation minima by mapping results to the convergent angle with the highest risk level.

In TMA, all aircraft continuously change their altitudes, whether climbing for departure or descending for arrival. When the relative position of two aircraft at CPA infringes both vertical and separation minima, the controller will issue heading-altering or speed-adjusting instructions to the aircraft without the right-of-way to achieve horizontal separation minima or altitude-changing instructions to achieve vertical separation minima.

For horizontal separation minimum assessment, we analyze a scenario where the non-right-of-way aircraft executes a collision avoidance maneuver, either adjusting its heading or reducing speed, at the candidate separation to ensure compliance with the WC threshold at the updated CPA. The speed reduction is prioritized, subject to the aircraft's operational speed envelope, for conflict resolution. Similarly, for vertical separation minimum assessment, we assume the non-right-of-way aircraft receives an ATC instruction to climb or descend to the specified vertical separation distance, ensuring no WC threshold violation occurs at the updated CPA.

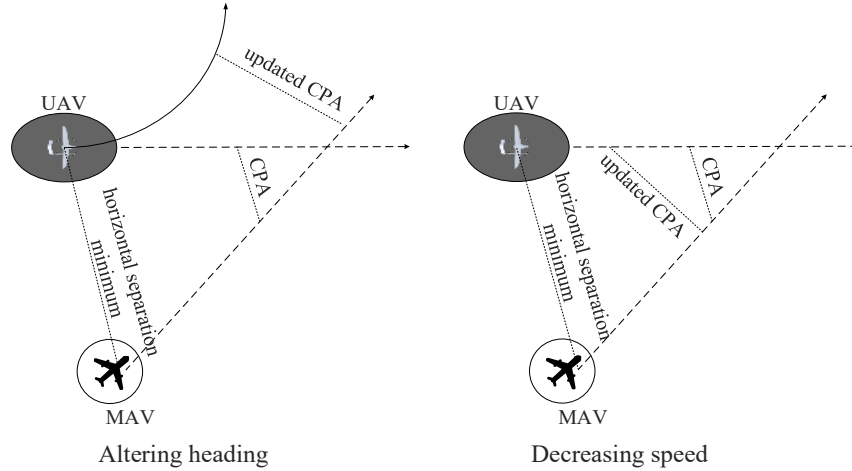


Fig. 4 The scenarios for horizontal separation minimum assessment

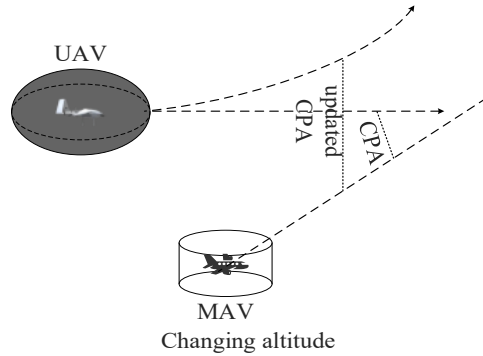


Fig. 5 The scenarios for vertical separation minimum assessment

iii. Right of way

The following hierarchical rules govern right-of-way assignments:

- (a) the UAV shall give way to the MAV in a normal encounter;
- (b) the MAV shall give way to the UAV in an emergent encounter;
- (c) the arrival shall give way to the departure, irrespective of UAV or MAV;
- (d) the arrival at a higher altitude shall give way to the arrival at a lower altitude;
- (e) the arrival further to the final approach fix (FAF) shall give way to the arrival closer to the FAF.

B. Simulation parameters setting

This simulation employs the B737-800 as the MAV representative and the TB001 (a large fixed-wing UAV) as the UAV representative. The arrival/departure traffic follows a Poisson process, with traffic density ranging from 30 to 50 aircraft per hour, and a fixed mix ratio of 1:4 between the number of UAVs and MAVs, reflecting the current operational pattern in Zigong TMA.

Table 1 Parameter settings

	MAV(B737-400)	UAV(TB 001)
Size (Length×Width×Height)	39.47m × 35.79m × 12.55m	10m × 20m × 3.3m
Trajectory conformity (TC)	Horizontal TC:	Horizontal TC:
	σ : 92.25 m	σ : 92.25m
	Vertical TC:	Vertical TC:
	σ : 15m	σ : 15m
Descend/ climb Speed in True Airspeed (TAS)	100~130m/s	83.3m/s
Rate of climb/Rate of descent	9m/s	10m/s
Deceleration in normal situations	0.098~0.49 m/s ²	1m/s ²
& emergent situations	0.98-1.96m/s ²	4m/s ²
Maximum pitch angle for ECA maneuver	-	±15°
DAA failure rate	-	0.18

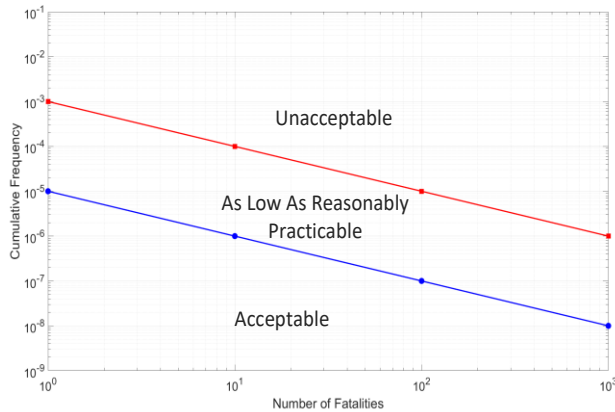
The latency time of the Monte Carlo simulation, primarily corresponding to the tactical conflict resolution phase, is critical for determining the WC threshold in integrated MAV-UAV operations. Key components of this latency—including surveillance, communication, controller and pilot response, as well as UAV total system maneuver delays—are referenced from NASA experimental studies [37] (see Tab. 2).

Table 2 Latency Time

	Surveillance	Communication	ATC	Pilot	UAV
Time (s)	2	6.77	16.7	18.95	25.07

C. Results and analysis

1) ELOS

**Fig. 6 Social Acceptable Risk Level in China**

(State Administration of Work Safety. Announcement No. 13 of 2013: Approval and issuance of 26 industry safety standards for production. Beijing: China Coal Industry Publishing House, 2013-06-08.)

Table 3 Parameters of ELOS Calculation

MAV weight (ton)	The speed of MAV (km/h)	Number of passengers onboard MAV	UAV weight (ton)	UAV speed (km/h)	Social influence ratios
79	463	190	3.2	320	97.6
50	407	10	3.2	260	65.8

The social influence ratios were determined according to Announcement No. 13 (2013) issued by China's State Administration of Work Safety (SAWS), as illustrated in Figure 6. To ensure comprehensive coverage of operational scenarios, the parameters for MAVs and UAVs used to calculate the ELOS were configured to represent the full performance envelope of both candidate aircraft in our simulation, detailed in Tab. 3. The weight range for the MAV was defined from the operational empty weight (OEW) to the maximum takeoff weight (MTOW) of the B737-800, accounting for its payload capacity. This approach ensures coverage of the widest possible ELOS range. The speeds of both MAV and UAV were selected based on their representative operational speed envelopes, typical for TMA operations. With reference to ICAO's TLOS for mid-air collision risk (1×10^{-9} for manned aviation) [50], the calculated ELOS range for MAV/UAV integrated operations in TMAs is 1.53×10^{-6} to 3.37×10^{-6} . To align with conservative risk management practices, this study adopts a baseline ELOS threshold of 1×10^{-6} .

2) WC Thresholds

To compute the WC thresholds, we first establish the NMAC boundaries, defined by a three-dimensional overlap of aircraft collision volumes corresponding to an acceptable MAC probability threshold of 0.1%. We analyze conflict scenarios involving two aircraft approaching at varying speeds, with crossing angles ranging from 30° to 150° in increments of 30° . As illustrated in Figure 7, the WC threshold exhibits a strong dependence on both relative speed and crossing angle: increasing either parameter results in larger WC thresholds.

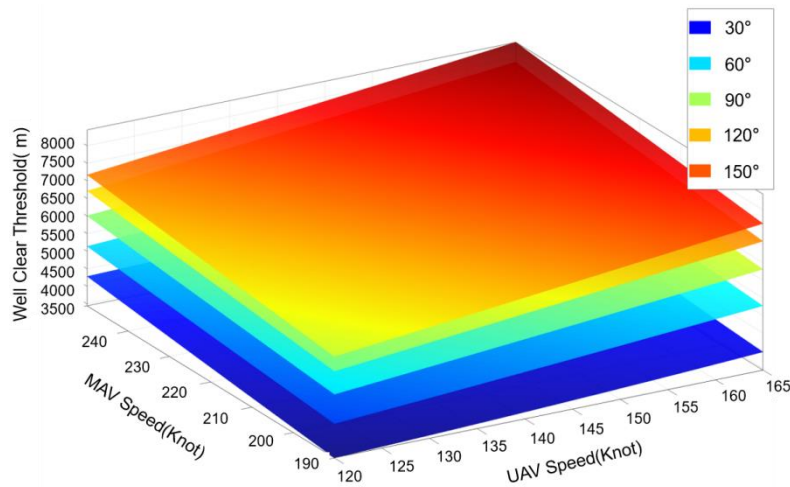
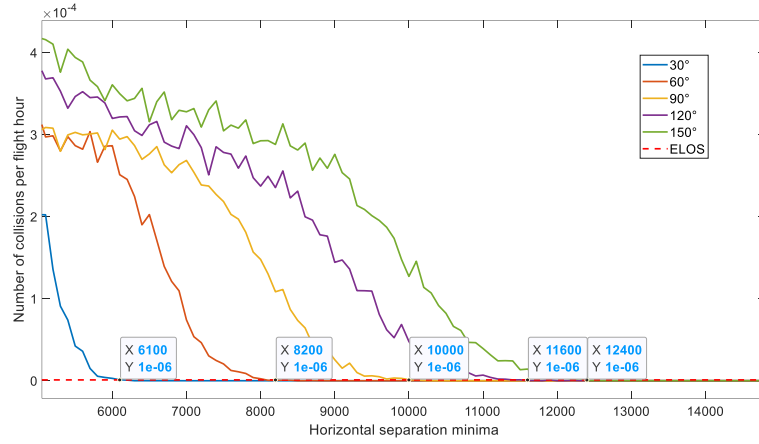


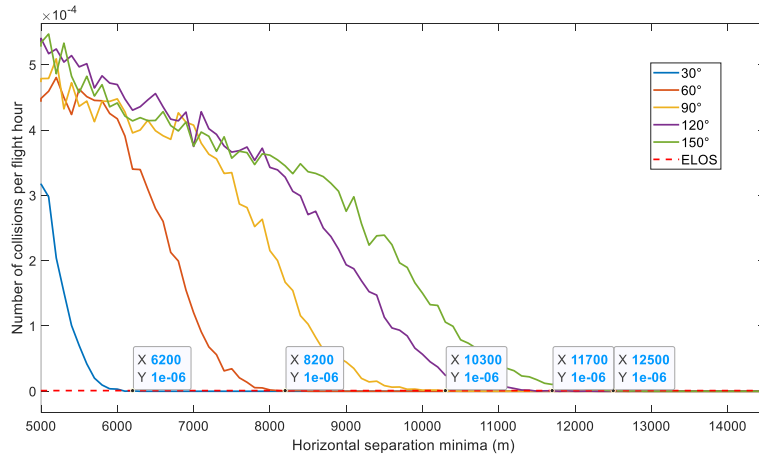
Fig. 7 Well Clear distance under different speeds and crossing angles

2) Horizontal Separation Assessment

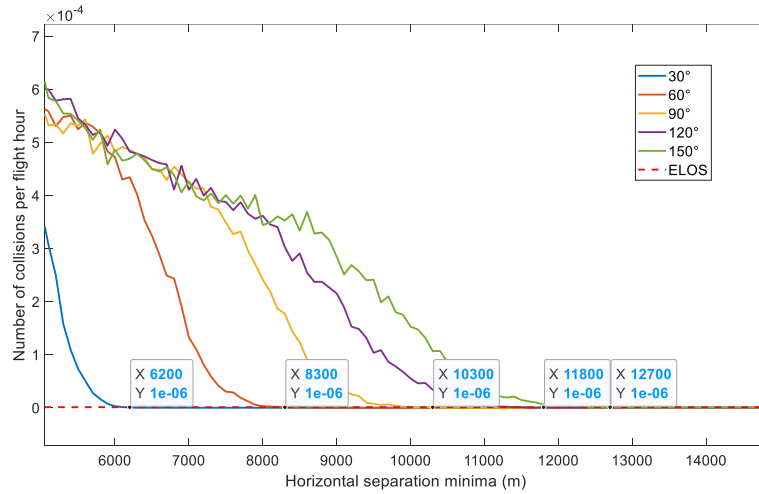
Figure 8 presents simulation results for assessing horizontal separation minima at encounter rates ranging from 30 to 50 per hour, with a fixed UAV/MAV ratio of 1:4. The collision risk decreases with increasing horizontal separation minima, as expected. The separation meeting the ELOS of 1×10^{-6} increases with the crossing angle between the conflict pair's trajectories. A larger crossing angle results in a greater relative speed, thereby increasing the likelihood of collisions. However, the collision risk doesn't have a significant positive relationship with the traffic density, contrary to initial expectations. The simulation results for horizontal separation minima assessment yield the same conclusion, regardless of the varying ratios of UAVs to MAVs, with a fixed number of 30 aircraft, as demonstrated in Figure 9. The most significant difference in horizontal separation minima is due to variation in MAV speeds: higher MAV speeds yield larger horizontal separation minima (see Figure 10).



(a) 30 aircraft



(b) 40 aircraft



(c) 50 aircraft

Fig. 8 Simulation results for horizontal separation minima assessment with varying encounter numbers increasing from 30 to 50 per hour with a fixed UAV/MAV ratio of 1:4

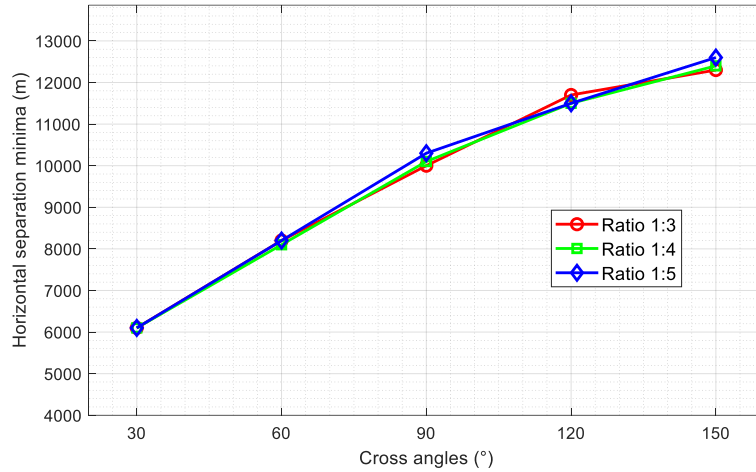


Fig. 9 Simulation results for horizontal separation minima assessment with varying UAV/MAV ratios at a fixed traffic density of 30 aircraft per hour

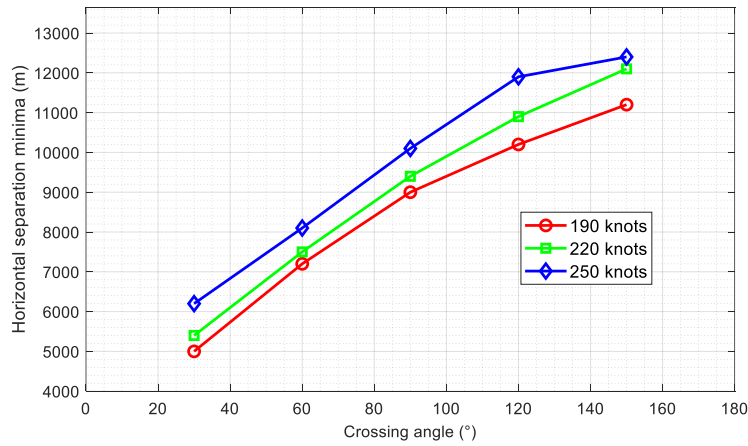


Fig. 10 Simulation results for horizontal separation minima assessment with varying speeds of MAV from 190 knots to 250 knots with 6 UAVs and 24 MAVs per hour

3) Vertical Separation Assessment

Figure 11 presents simulation results evaluating vertical separation minima under conditions of 40 encounters per hour on reciprocal tracks, with varying UAV-to-MAV ratios from 1:3 to 1:5. Consistent with horizontal separation trends, collision risk shows a positive relationship with both traffic density and the fraction of UAV numbers. Notably, aircraft speed emerges as the dominant factor influencing separation minima, overshadowing other collision-risk variables. A secondary observation (Figure 12) reveals no positive correlation between crossing angles and collision risk, suggesting angular variations have a negligible impact under these modeled conditions. The separation distance required to meet the European Target Level of Safety (ETLOS) of 1×10^{-6} falls within the range of 678 to 707m, substantially exceeding the conventional 300m vertical separation minimum for MAV operation.

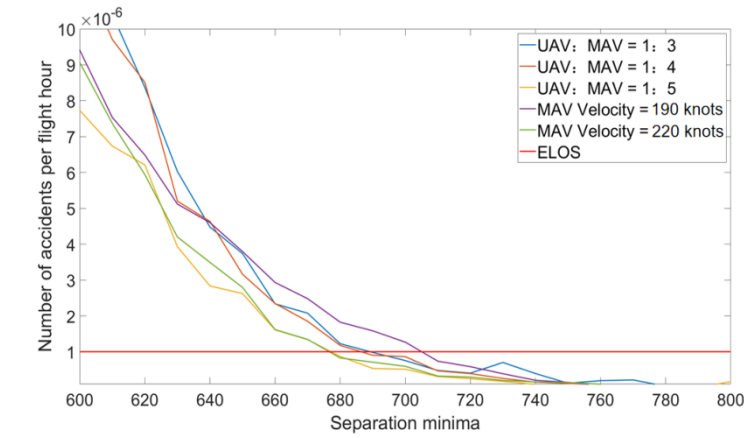


Fig. 11 Vertical separation standard assessment for flying on a reciprocal track

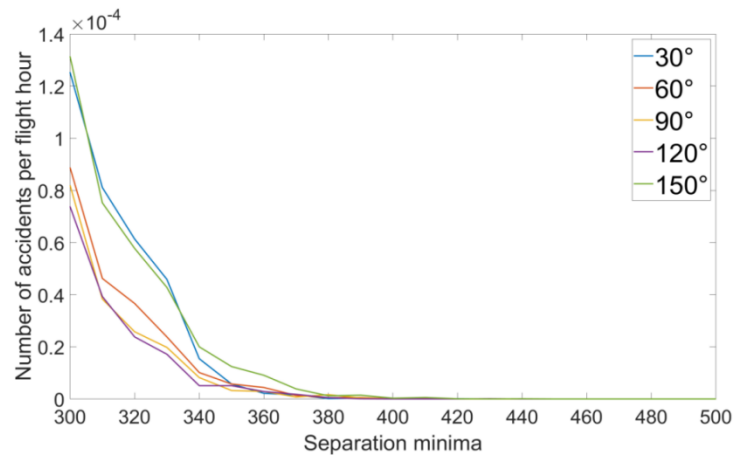


Fig. 12 Vertical separation standard assessment for different trajectory crossing angles

4) Convergence Analysis of Simulations

Monte Carlo simulations inherently involve statistical uncertainty, necessitating a large number of experimental runs to ensure convergence of the results. Figure 13 illustrates the variation in accident frequency per unit hour as the number of experiments increases from 1,000 to 8,000. The results demonstrate asymptotic stabilization beyond approximately $N = 5,000$ simulations. To balance computational efficiency with statistical reliability, this study adopts $N = 5,000$ simulations for all subsequent analyses.

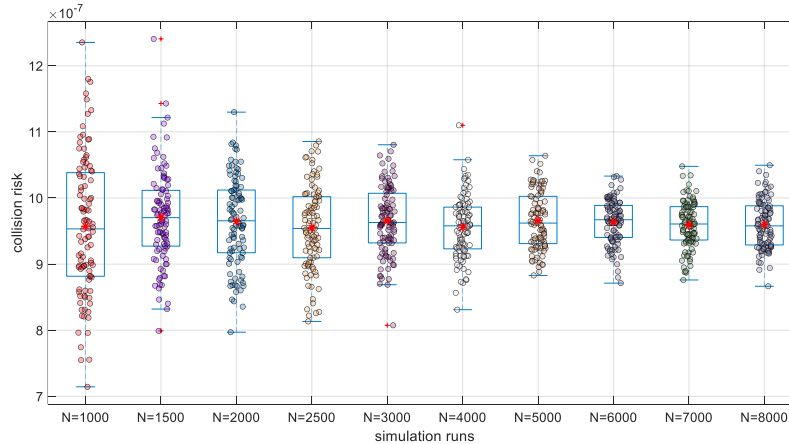


Fig. 13 Relationship between the number of accidents per flight hour and the Monte-Carlo iterations

V. Conclusion

This study investigates the integration of UAVs into terminal airspace, analyzing three key mitigation phases: strategic conflict management, tactical deconfliction, and reactive collision avoidance. We propose a trajectory-conformity-based probabilistic analysis framework for evaluating collision risk across interdependent subsystems during encounters between UAVs and MAVs. From a regulatory perspective, we introduce a novel ELOS metric, which systematically determines separation requirements by integrating UAV-specific kinetic energy profiles and societally acceptable risk thresholds.

Using Monte Carlo simulations, we conduct a comprehensive safety assessment under multiple operational scenarios, examining the effects of traffic density variations, UAV speeds and UAV-to-MAV mixing ratios. The results establish scenario-specific horizontal and vertical separation minima optimized for UAV-MAV integrated operation in terminal airspace. The horizontal separation minima evaluation results indicate a systematic increase in required separation minima with crossing angles, MAV speeds and traffic densities. However, the vertical separation minimum evaluation results do not show a positive relationship with the crossing angles. Subsequent analysis of traffic density effects reveals a direct correlation between higher traffic densities and greater separation requirements.

In conclusion, the primary contributions in this research include:

- i. Building of a high-fidelity analysis and simulation framework in determining the separation minima; and
- ii. Proposing an ELOS quick-estimated model incorporating the kinetic energy proportionality principle and societal risk acceptance threshold for UAV-MAV integrated operations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the National Key Research and Development Program Project of China (NO. 2022YFB4300904) and the Joint Funds of the National Natural Science Foundation of China (No. U2433214).

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